

Beam-induced depolarization of the HJET target: corrected analysis and holding-field windows for RHIC and the EIC

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Abstract

We present the corrected analysis of beam-induced depolarization of the polarized atomic hydrogen jet target (HJET) under EIC beam conditions, superseding the quantitative results of Sec. VI.D of Ref. [1]. With the corrected harmonic amplitudes of the bunch-train field, only the two low-frequency transitions f_{12} and f_{34} can be driven resonantly. Folding the static holding-field distribution along the atomic trajectories into the dynamics converts each resonance into an exclusion corridor whose width is set by the field excursion over the drive region; between the corridors, wide acceptable windows remain. For the ~~RHIC~~ magnet the corrected depolarization at the operating point is $\Delta Q/Q \approx 0.012\%$ at 120 mT, robust against the atomic velocity, and an unconditional window extends from about 247 mT to the f_{34} , $n = 8$ crossing at 1177 mT, with the total-transfer minimum near 530 mT. For the EIC, where compact HTS guide-field coil pairs in three Cartesian orientations replace the resistive magnet and an RF shield confines the bunch fields, we compute the corridor maps per orientation from the 3D field calculations: the B_z pair matches the RHIC magnet at the 400 mT operating point, the B_y orientation is safe but ceiling-limited by the f_{34} , $n = 8$ crossing at 1177 mT, the B_x orientation is cleanest above 496 mT, and the common acceptable window across all orientations is 325 mT and above, making elevated-field operation mandatory for the EIC system.

1 Introduction

The absolute calibration of the proton beam polarization at RHIC, and in the future at the EIC, rests on elastic scattering off the polarized atomic hydrogen gas jet target (HJET). Under EIC beam conditions, with an order of magnitude shorter bunches and higher current, the time-dependent magnetic field of the passing bunches can drive transitions between the hyperfine states of the target atoms and thereby depolarize the target. Reference [1] analyzed this mechanism and recommended an elevated holding field. Its harmonic-amplitude normalization was subsequently found to be in error [2]: the amplitudes of Sec. VI.D of Ref. [1] represent the single-bunch spectral envelope normalized to the peak field and omit the Fourier normalization of the periodic bunch train, overestimating the Rabi frequencies by a factor of about 60. An Erratum is in preparation; its working draft is Appendix C.

This note contains the corrected analysis, developed from first principles and self-contained. Beyond the correction it adds the element that a point evaluation at fixed field cannot capture: the static holding field varies along the atomic trajectories, so the resonance condition is met locally, and each resonance widens into an exclusion corridor whose width is a property of the magnet, not of the resonance dynamics. On this basis we map the acceptable holding-field windows for the existing RHIC magnet and, using the 3D field calculations of the planned compact HTS guide-field coils and the RF shield, for the three field orientations foreseen at

the EIC. This note is the analysis record; the bulk of the material is intended for a dedicated publication.

2 Theoretical framework

The notation follows Ref. [1] throughout; equation numbers in parentheses after “Ref.” refer to that paper. ~~Subscripts are named at first use.~~

2.1 Hyperfine structure in the holding field

The ground state of atomic hydrogen in a static field B splits into the four hyperfine states $|1\rangle \dots |4\rangle$ of Eq. (5) of Ref. [1], with energies given by the Breit–Rabi formula, Eq. (11) there, in terms of the dimensionless field

$$x = \frac{B}{B_c}, \quad B_c = \frac{E_{\text{hfs}}}{g_J \mu_B} = 50.684 \text{ mT}, \quad (1)$$

Eqs. (7) and (10) of Ref. [1], where $E_{\text{hfs}} = h f_{\text{hfs}}$ and $f_{\text{hfs}} = 1420.405\,748$ MHz is the zero-field hyperfine splitting. The transition frequencies $f_{ij} = (E_i - E_j)/h$, Eq. (13) there, that matter below are the two that change the nuclear spin and are reached by the bunch harmonics,

$$f_{12}(B) = \frac{1}{2} f_{\text{hfs}} \left(1 + x - \sqrt{1 + x^2} \right), \quad f_{34}(B) = \frac{1}{2} f_{\text{hfs}} \left(1 - x + \sqrt{1 + x^2} \right), \quad (2)$$

with $f_{12} \rightarrow f_{\text{hfs}}/2$ from below and $f_{34} \rightarrow f_{\text{hfs}}/2$ from above as B grows. Their transition strengths are the matrix elements $M_{12} = M_{34} = \sin^2 \theta$ of Eq. (C4) of Ref. [1], with the mixing angle θ of the $|2\rangle, |4\rangle$ states defined by $\tan(2\theta) = 1/x$, Eq. (B7) there. The Rabi frequency below carries the square root of the transition strength,

$$\sqrt{M_{12}} = \sqrt{M_{34}} = \sin \theta, \quad \theta = \frac{1}{2} \arctan(1/x), \quad (3)$$

which decreases from $\sin(\pi/4)$ at low field as the electron and proton spins decouple.

2.2 The beam-induced field

A round Gaussian beam of rms radius σ_r carrying the bunch current $I_b(t)$ produces the azimuthal field of Eqs. (44) and (45) of Ref. [1],

$$B(r, t) = \frac{\mu_0}{2\pi r} I_b(t) F(r), \quad F(r) = 1 - \exp\left(-\frac{r^2}{2\sigma_r^2}\right), \quad (4)$$

and we write $B_{\text{pk}}(r) = \mu_0 I_b^{\text{pk}} F(r)/(2\pi r)$ for its value at the bunch peak (subscript pk), with $I_b^{\text{pk}} = Q_b/(\sqrt{2\pi} \sigma_t)$ the peak bunch current of Table II there. For the EIC flattop parameters ($\sigma_r = 0.656$ mm, $I_b^{\text{pk}} = 22.0$ A) B_{pk} reaches 3.03 mT at $r = 1.04$ mm, Fig. 15(b) of Ref. [1].

2.3 Spectrum of the bunch train

The single bunch is the Gaussian current of Eq. (14) of Ref. [1], with Fourier transform $\tilde{I}_b(f) = I_b^{\text{pk}} \exp(-2\pi^2 f^2 \sigma_t^2)$, Eq. (18) there, and the train is its convolution with the comb of spacing $\tau_b = 1/f_b$, whose spectrum is $\tilde{I}(f) = \tilde{I}_b(f) \sum_n \delta(f - n f_b)$, Eq. (19) there. Written as a real Fourier series in time, the same statement reads

$$I(t) = f_b \tilde{I}_b(0) + 2f_b \sum_{n=1}^{\infty} \tilde{I}_b(n f_b) \cos(2\pi n f_b t), \quad f_b \tilde{I}_b(0) = \sqrt{2\pi} \sigma_t f_b I_b^{\text{pk}} = I_{\text{avg}}, \quad (5)$$

and the zeroth term reproduces the average current I_{avg} of Table II of Ref. [1], which fixes the normalization. The oscillation amplitude of harmonic n of the induced field at radius r follows by inserting Eq. (5) into Eq. (4),

$$B_1^{(n)}(r) = \frac{2f_b \tilde{I}_b(nf_b)}{I_b^{\text{pk}}} B_{\text{pk}}(r) = 2\sqrt{2\pi} \sigma_t f_b B_{\text{pk}}(r) e^{-2\pi^2 n^2 f_b^2 \sigma_t^2}, \quad (6)$$

the product of the duty factor ($\sqrt{2\pi} \sigma_t f_b = 0.0455$ at the EIC flattop), the factor 2 of a real series, and the Gaussian envelope. The subscript 1 is that of the rf field B_1 of Ref. [1]; Eq. (6) is the quantity that was overestimated there, where the envelope normalized to the peak field, without the factor $2f_b/I_b^{\text{pk}}$, was used in Eq. (34) and Sec. VI.D. With $\sigma_t = 0.2$ ns and $f_b = 90.69$ MHz the envelope confines appreciable amplitudes to $n \lesssim 12$, so only harmonics that can meet f_{12} or f_{34} below about 1.2 T matter; the crossing fields follow from Eq. (2) and are listed with the maps below.

The Rabi frequency (subscript R) of a transition driven at harmonic n is the precession frequency of the electron spin in the rotating component of $B_1^{(n)}$, reduced by the transition strength,

$$\omega_R = \frac{1}{2} \gamma_H B_1^{(n)}(r) \sin \theta, \quad \gamma_H/2\pi = 28.025 \text{ GHz/T}, \quad (7)$$

with γ_H the gyromagnetic ratio of Table I of Ref. [1]; the factor $\frac{1}{2}$ belongs to the full-amplitude convention of Eq. (6) (a linearly oscillating field of amplitude B_1 is two counter-rotating fields of amplitude $B_1/2$, of which one drives the transition). Equation (7) replaces Eq. (34) of Ref. [1], $f_{\text{Rabi}} = \gamma_H B_1$, which is the correction detailed in Appendix C.

2.4 Two-level dynamics and the three response regimes

Along a trajectory both $\omega_R(t)$ (through the distance $r(t)$ from the ion beam) and the detuning $\Delta(t) = 2\pi f_{ij}(B(t)) - 2\pi n f_b$ (through the static field $B(t)$ along the path) are time dependent, and the two-level Schrödinger equation for the pair of states i, j is integrated numerically (piecewise-constant propagator, 2×10^4 steps per traversal). The transfer probability w_{ij} into the other state of the pair is the quantity of interest (Sec. 2.6); τ denotes the time an atom spends in the drive region and v its velocity.

The numerical result alone does not say what it depends on. The purpose of the classification below is to attach to every feature of the maps the inputs it is sensitive to, so that it is clear which numbers are robust and which need a measured input. Three limiting regimes do this: the time-averaged tails depend only on the drive amplitude and the detuning, not on the atomic velocity, the transit time or the shape of the static field along the path, and they set the valley floors and the 10^{-3} corridor edges, which is why those survive every sensitivity test of this note (velocity, field model, drive-region boundary); the transit-limited response depends on the velocity as $1/v^2$ and lives only within a few millitesla of a crossing, where no operating point is placed; the resonance traversals depend on the velocity as $1/v$ and on the static field model, since it is the field variation along the path that carries the atom through the resonance, and they set the interior of the corridors and the high-field structure of the EIC B_y orientation. What the map quotes at an operating point therefore needs the velocity only loosely (Sec. 2.8) and the magnet model only through its excursion over the drive region (Sec. 2.5), and what it quotes inside a corridor would need both precisely, which is the reason no operating point is quoted there.

1. Time-averaged tails, $\Delta\tau \gg 1$. For constant parameters the Rabi formula gives $w = (\omega_R^2/\Omega^2) \sin^2(\Omega t/2)$ with the generalized Rabi frequency $\Omega = \sqrt{\Delta^2 + \omega_R^2}$, the precession frequency of the two-level system about the effective field of the detuning and the drive. Far off resonance, many oscillation periods fit into the transit, the $\sin^2$ factor averages

far from the
resonant max?

use an introductory eq. that defines $\Delta\tau$

are we showing maps for this velocity?

(not shown)

to $\frac{1}{2}$, and $w \simeq \frac{1}{2}(\omega_R/\Delta)^2$: independent of the interaction time and hence of the atomic velocity. The valley floors of all maps below, and the corridor widths at the 10^{-3} contour, live in this regime; the computed maps at $v = 1562$ and 1807 m/s indeed yield identical corridor edges and operating-point values within the ensemble statistics.

2. *Transit-limited response*, $\Delta\tau \lesssim 1$. Close to resonance the phase Ωt does not complete a period within the transit and the same formula reduces to $w \simeq \sin^2(\omega_{RT}/2) \approx (\omega_{RT}/2)^2 \propto 1/v^2$, growing with the interaction time $\tau = 2L_{\text{drive}}/v$; relevant only in the immediate wings of a crossing.

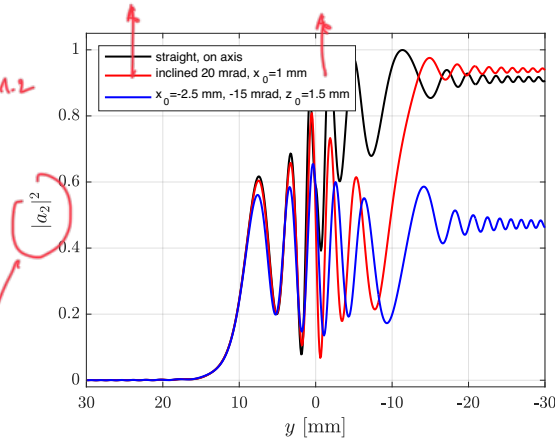
3. *Resonance traversals* (Landau-Zener). When the static field variation along the path carries the atom through the resonance, the detuning $\Delta(t)$ changes sign, and for a linear sweep the transfer per traversal is the Landau-Zener probability $w = 1 - \exp(-\pi\omega_R^2/2|\dot{\Delta}|) \simeq \pi\omega_R^2/(2|\dot{\Delta}|)$ [3, 4], with $|\dot{\Delta}| \propto v|dB/dy|$, hence $w \propto 1/v$; this governs the interior of the corridors and, at the EIC, the B_y -orientation traversals discussed in Sec. 4.

why is only B_y mentioned here.

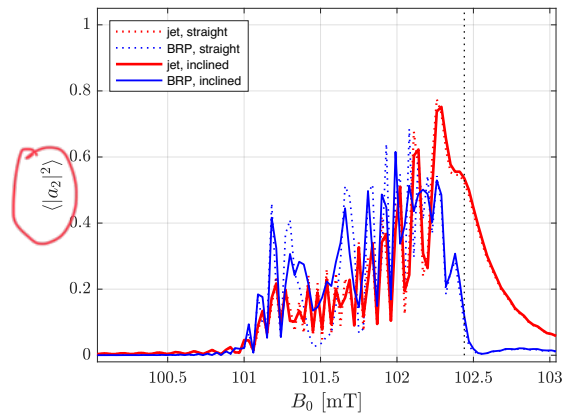
Figure 1 shows the three regimes at work in one crossing, f_{12} at $n = 6$ (102.44 mT) in the RHIC magnet. Figure 1a follows single atoms through the drive region at a setpoint 0.35 mT below the crossing: the population of the target state builds up while the atom traverses the resonance where the static field along its path reaches B_x , oscillates at the local Rabi frequency, and settles to a permanent value when the drive falls off; the three tracks differ in offset and inclination and therefore in where they cross and how strongly they are driven. Figure 1b scans the setpoint across the crossing for an ensemble of 60 atoms and records the transfer at the jet plane and at the exit (BRP): inside the intrinsic band both are of order unity and structured by the traversal phases; below the band the jet-plane value exceeds the exit value, because the adiabatic dressing of the time-averaged tails is present while the atom is in the field and reverts afterwards; above the crossing the field along the path never reaches B_x , the exit value collapses and only the jet-plane tail remains. Straight and inclined ensembles differ little, the statement of Sec. 2.8 that inclinations are a minor input.

nuclear transfer is done and vice versa is!

expand the y-scale to 1.2



(a) Population of the target state along the path for three single atoms at $B_0 = 102.09$ mT: on axis, inclined by 20 mrad at $x_0 = 1$ mm, and offset by $x_0 = -2.5$ mm, $z_0 = 1.5$ mm with -15 mrad.



(b) Ensemble average of 60 atoms against the setpoint, at the jet plane (red) and at the exit (BRP, blue), for straight (dotted) and inclined (solid) tracks; the vertical line marks B_x .

Figure 1: Passage through the f_{12} , $n = 6$ crossing at 102.44 mT in the RHIC magnet with the static field of Eq. (16), $v = 1807$ m/s. Script: lowharm_passage_v10.m.

this is not defined. Should be properly introduced.

why not 1506 m/s?

This is what is used later!

2.5 The corridor mechanism

The holding field is not uniform: along the flight path through the drive region the static field varies by an excursion δB_{exc} (for the RHIC magnet, $\delta B_{\text{exc}} = a_y L^2 = 1.68 \text{ mT}$ over the $\pm 30 \text{ mm}$ chamber region). An atom is resonantly transferred if *any point* of its path satisfies the resonance condition, so each crossing field $B_{\times}$ broadens into an exclusion corridor of intrinsic width δB_{exc} , with a sharp edge at $B_{\times}$ itself and a soft edge on the side toward which the field along the path deviates, i.e. on the side given by the sign of the curvature. Two widths must be distinguished. The intrinsic width, where the transfer is of order unity, is a property of the static field (and, at the EIC, of the RF shield that bounds the drive region), not of the resonance dynamics; this is the element absent from any fixed-field evaluation. The 10^{-3} corridors drawn in the maps are 10 to 20 mT wide and are set by the Landau-Zener and tail contributions of the on-axis dynamics; at that level the 1.7 mT intrinsic width of the RHIC magnet, and with it the sign of its curvature, is not resolved (Sec. 3). Figure 2 shows both widths for the same crossing as Fig. 1, on a logarithmic scale over $\pm 15 \text{ mT}$: the intrinsic band of 1.34 mT below $B_{\times}$ (the axial rise of Eq. (16) at the crossing field) where the transfer is of order unity, and the 10^{-3} corridor from 97.3 to 107.5 mT set by the tails, 5 mT on either side of the crossing, four times the intrinsic band. In the wings the exit (BRP) curve lies two orders of magnitude below the jet-plane curve, the reversion of the adiabatic dressing that Sec. 2.7 uses. At the RHIC setpoint of 120 mT the jet-plane transfer from this crossing alone is about 10^{-4} ; the 1.22×10^{-4} of the map is the sum over all crossings.

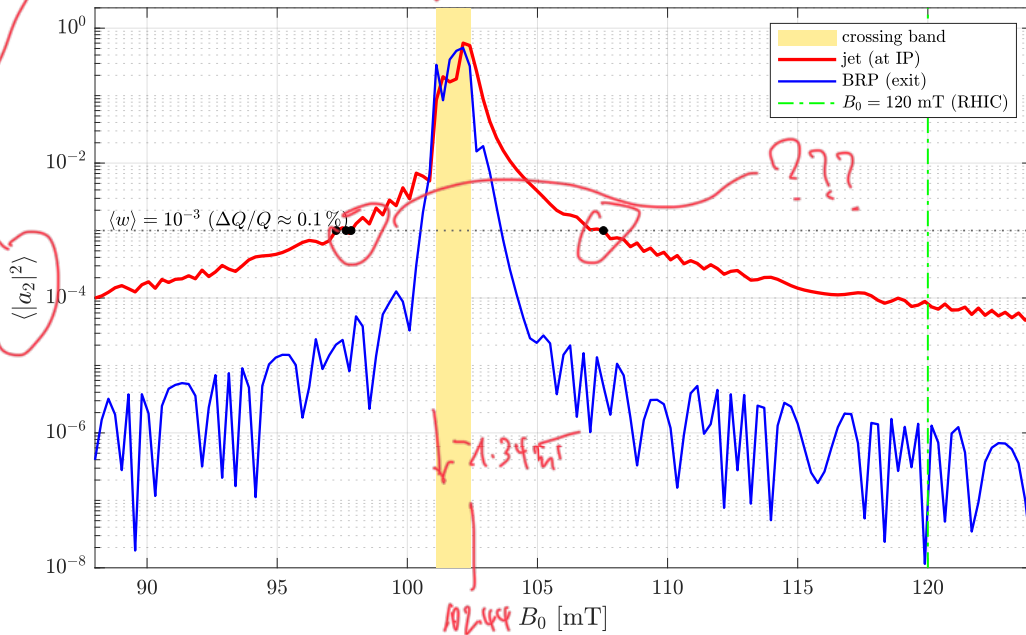


Figure 2: Clearance around the f_{12} , $n = 6$ crossing at 102.44 mT in the RHIC magnet with the static field of Eq. (16): ensemble-averaged transfer at the jet plane (red) and at the exit (BRP, blue) against the setpoint, on a logarithmic scale. The shaded band is the intrinsic corridor width $\delta B_{\text{exc}} = 1.34 \text{ mT}$; the dotted line and the markers give the 10^{-3} criterion and its edges at 97.3 and 107.5 mT; the dash-dotted line marks the RHIC setpoint. Script: `clearance_scan_v7.m`.

2.6 Observable

Let w_{ij} denote the ensemble-averaged probability that an atom entering the interaction region in hyperfine state $|i\rangle$ leaves it in state $|j\rangle$ after one traversal, computed from the dynamics above;

the pairs $|1\rangle \leftrightarrow |2\rangle$ and $|3\rangle \leftrightarrow |4\rangle$ are the only ones coupled by the accessible harmonics. The jet operates with the injected hyperfine pairs $\{|1\rangle, |4\rangle\}$ and $\{|2\rangle, |3\rangle\}$. To leading order the loss of the measured nuclear polarization Q is

$$\frac{\Delta Q}{Q} = -(w_{12} + w_{34}) \quad (8)$$

for both injected combinations alike: atoms are not lost, their polarization is. All maps below show $\langle w \rangle = \langle w_{12} + w_{34} \rangle$ averaged over the trajectory ensemble. As acceptance criterion we require that the systematic error on the target polarization from beam-induced depolarization stay below 0.1%, so that it remains subdominant in the polarimetry error budget; this translates into $\langle w \rangle < 10^{-3}$, the contour drawn in all maps.

2.7 What the maps mean for the polarimeter

The maps show the transfer averaged over the trajectory ensemble and recorded at the jet plane. The polarimeter does not measure that quantity, and this subsection states what it measures instead and how the two are related.

The beam polarization is obtained from the two asymmetries (Eq. (4) of Ref. [1]),

$$P_b = \frac{\epsilon_b}{\epsilon_t} Q_{\text{BRP}}, \quad \epsilon_b = A_y P_b, \quad \epsilon_t = A_y Q_{\text{eff}}. \quad (9)$$

Here ϵ_b is the beam-spin asymmetry averaged over the target spin states and ϵ_t the target-spin asymmetry averaged over the beam spin states. The analyzing power A_y cancels in the ratio. Two different target polarizations enter, however. ϵ_t is produced by the atoms that actually scatter, so Q_{eff} is the polarization of the atoms inside the hadron beam at the moment of scattering: the jet-plane polarization weighted by the beam density. Q_{BRP} is the polarization of the whole atomic beam after it has left the interaction region, unweighted. Inserting Eq. (9) into itself,

$$\frac{P_b^{\text{reported}}}{P_b} = \frac{Q_{\text{BRP}}}{Q_{\text{eff}}} \approx 1 + w_{\text{jet}}^w - w_{\text{BRP}}, \quad (10)$$

where w_{jet}^w is the beam-weighted transfer at the jet plane and w_{BRP} the unweighted transfer after the full passage, both small. The systematic error of the polarimeter is the difference of the two averages, not either of them, and the ensemble average $\langle w \rangle$ of the maps is a third quantity that bounds both. Three effects separate them.

2.7.1 Half against full passage

w_{jet} is taken halfway through the drive region, w_{BRP} at its end. In the time-averaged-tail regime of Sec. 2.4 the transfer is the admixture of the dressed state, $w = \frac{1}{2}(\omega_R/\Delta)^2$, which follows the local drive adiabatically: it is present while the atom is in the bunch field and reverts to zero when the atom leaves it. w_{BRP} therefore retains only the non-adiabatic remainder, which Figs. 1b and 2 show to lie well below the jet-plane value in the tails. For resonance traversals, by contrast, the transfer is permanent and the BRP sees two traversals where the jet plane has seen one, $w_{\text{BRP}} \approx 2 w_{\text{jet}}$.

2.7.2 Which atoms are sampled

The ensemble of Sec. 2.8 has transverse offsets of 2.6 mm rms; the recoil protons come from the atoms inside the hadron beam, whose density is Gaussian with $\sigma_x = 1.61$ mm and $\sigma_y = 0.27$ mm at flattop (Table II of Ref. [1]). In the tail regime w is local and proportional to ω_R^2 at the scattering point, so the weighted floor is

$$w_{\text{jet}}^w \propto \langle B_{\perp}^2 \rangle_{\text{beam}} = \int \rho(x, y) B_{\perp}^2(x, y) dx dy, \quad (11)$$

with ρ the beam density and $B_\perp$ the component of the bunch field perpendicular to the holding field, against $\langle B_{\text{pk}}^2(|x_0|) \rangle$ over the ensemble at $y = 0$ for the maps.

2.7.3 Which component drives

Only the component of the bunch field perpendicular to $\vec{B}_0$ drives the $\Delta m_F = \pm 1$ transitions f_{12} and f_{34} . The maps use the full magnitude $|B|$ for every orientation. For the B_z pair (holding field along the ion beam) the azimuthal bunch field is entirely perpendicular and $B_\perp = |B|$. For the B_y pair and the RHIC magnet (holding field along the atomic path) $B_\perp = B_x$; for the B_x pair $B_\perp = B_y$. Inside a Gaussian beam the field is linear in the coordinates,

$$B_x = -\frac{\mu_0 I}{2\pi} \frac{y}{\sigma_y(\sigma_x + \sigma_y)}, \quad B_y = +\frac{\mu_0 I}{2\pi} \frac{x}{\sigma_x(\sigma_x + \sigma_y)}, \quad (12)$$

(the linearized Bassetti-Erskine field), and weighting with the beam density gives $\langle x^2 \rangle = \sigma_x^2$, $\langle y^2 \rangle = \sigma_y^2$, hence

$$\langle B_x^2 \rangle_{\text{beam}} = \langle B_y^2 \rangle_{\text{beam}} = \left(\frac{\mu_0 I}{2\pi} \right)^2 \frac{1}{(\sigma_x + \sigma_y)^2} \quad (13)$$

for any aspect ratio. The ellipticity of the beam therefore favours no orientation; the atoms that scatter sit inside the beam, where the ribbon-like field of a flat beam has not yet formed. The intuitive argument that on the axis the azimuthal field is parallel to B_x and the B_x map is therefore conservative by two orders of magnitude is true for the axis alone and irrelevant after the weighting.

Evaluating Eq. (11) with the full field of the elliptical Gaussian beam gives (Table 1, script `beam_weighting_v1.m`) the ratio of the beam-weighted floor to the map floor. For the flattop beam the B_x and B_y pairs and the RHIC magnet are at 0.38, the B_z pair at 0.76; at injection the beam is 2.2 times larger in both directions while the maps keep the flattop beam, and the factors fall by a further factor 5. For a round beam of the same area the weighting alone would raise the floor by 1.5; the projection more than compensates.

Table 1: Ratio of the beam-weighted drive $\langle B_\perp^2 \rangle_{\text{beam}}$ at the scattering points to the map reference $\langle B_{\text{pk}}^2 \rangle$ (round beam, ensemble x_0 rms 2.6 mm, jet plane), per holding-field orientation; source `beam_weighting_v1.m`.

beam	σ_x [mm]	σ_y [mm]	B_z pair	B_y pair, RHIC	B_x pair
EIC flattop, IP4	1.61	0.27	0.76	0.38	0.38
round equivalent	0.66	0.66	1.51	0.77	0.74
EIC injection, IP4	3.51	0.69	0.15	0.08	0.08

Putting the three effects together for the floors: the systematic of Eq. (10) is dominated by w_{jet}^w , since w_{BRP} is the non-adiabatic remainder, and amounts to 0.4 to 0.8 of the map floor with the sign that the reported beam polarization is too high: about 10^{-4} at 120 mT on the RHIC magnet and below 10^{-5} above 300 mT for every orientation. For the corridor edges none of the three effects matters: resonance traversals occur where the static field along the path crosses B_x , at $|y|$ of millimetres to 30 mm, where the bunch field is azimuthal, the projection is unity for the relevant component and the beam weighting has nothing to select on. The maps are therefore conservative for the polarimetry by a factor 1.3 to 2.6 in the floors and exact in the edges. A run of the trajectory code with per-atom recording, beam weighting and the projected drive, to confirm these estimates inside the full dynamics, is the next step of the analysis.

2.8 Trajectory ensemble and atomic velocity

The transfer probability of Sec. 2.4 depends on the trajectory twice: through $r(t)$, the distance from the ion-beam axis along the trajectory, which enters the drive amplitude as $r = r(t)$ in Eq. (4), and through $B(t)$, the static field sampled along the path, which sets the detuning. The maps below therefore average over an ensemble representing the atomic beam at the interaction point.

Transverse phase space. The jet has a Gaussian transverse density of $\sigma \approx 2.6$ mm at the interaction point (measured profile, FWHM 5.8 mm [5]); the ensemble draws offsets from this distribution, truncated at 4.5 mm by the aperture system around the interaction region [6]. Track inclinations are drawn with an rms of 20 mrad, the scale set by the source divergence transmitted through the sextupole system; the analysis is insensitive to this value at the level of the quoted results, and the phase-space-resolved ensemble from the atomic-beam tracking effort will replace it in due course (Table 3).

Velocity, mean. The mean atomic velocity was measured by time of flight, with a fast chopper in the fourth pumping stage and a quadrupole mass analyzer at the interaction point, extrapolated to infinite chopper speed: 1562 ± 50 m/s at 75 K nozzle temperature [5]. The same measurements showed the mean to be stable within ± 50 m/s against dissociator parameters, because the transmitted spectrum is fixed by the sextupole system, as follows.

Velocity, spread. The spread at the interaction point was never measured, but it is bounded by the optics. A sextupole focuses low-field-seeking atoms through the potential $U = \mu_B B_p (r/r_p)^2$, i.e. it acts as a lens of focal length

$$f = \frac{mv_p^2 r_p^2}{2\mu_B B_p L}, \quad \text{derive this} \quad (14)$$

with B_p the pole-tip field, r_p the bore radius and L the magnet length: the optics is strongly chromatic, $f \propto v^2$, and the multi-magnet system images the nozzle onto the target region only for a band of velocities. Tracking the source distribution, the drifted Maxwellian of a supersonic expansion [7],

$$f(v) \propto v^3 \exp\left[-\frac{m(v-u)^2}{2k_B T_{\parallel}}\right], \quad (15)$$

with drift u at the 75 K terminal velocity and $T_{\parallel} = 20$ K, $T_{\perp} = 1.45$ K [5], through the documented six-magnet geometry and aperture set [6, 8] yields the acceptance function of Fig. 3: a transmitted spread of $12 \pm 3\%$ rms with a hard acceptance cutoff below about 1300 m/s, slow atoms being overfocused by Eq. (14) onto the tapered bores. The cutoff is protective, since the slow side is where the velocity-dependent regimes of Sec. 2.4 enhance the transfer.

By the regime structure, the velocity and its spread affect only the resonance-traversal and transit-limited contributions; the corridor edges and valley floors are velocity independent, as verified by direct computation.

2.9 Correspondence to Ref. [1]

For readers of the original paper, the equations of this section are those of Ref. [1] with one exception: Eq. (1) is Eqs. (7) and (10) there; Eq. (2) follows from the Breit–Rabi energies, Eqs. (11) and (13); Eq. (3) is the square root of the M_{12} , M_{34} entries of Eq. (C4) with θ from Eq. (B7); Eq. (4) is Eqs. (44) and (45); Eq. (5) is Eqs. (18) and (19) written as a real series; the beam parameters are those of Table II. The exception is the drive amplitude: Eqs. (6) and (7) replace the amplitude normalization underlying Sec. VI.D and Eq. (34), and the corrected treatment replacing Eqs. (C2) and (C3) is given in Appendix C. Sections III and IV of Ref. [1] are unaffected by the present analysis.

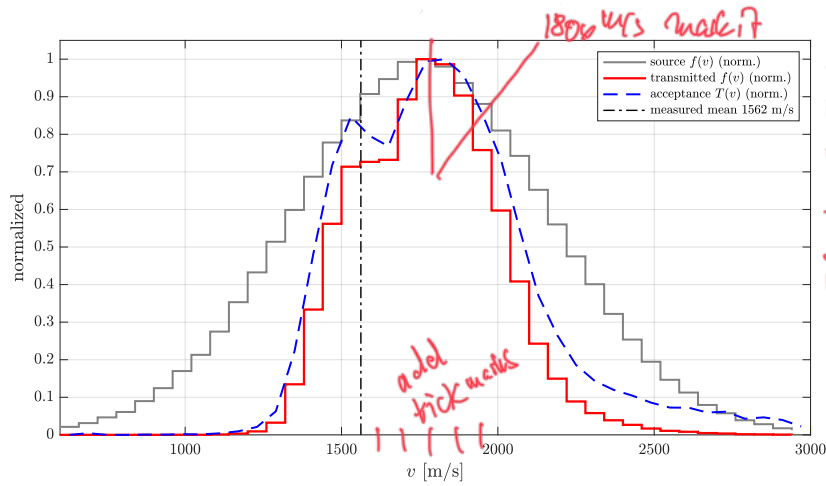


Figure 3: Velocity acceptance of the sextupole system from Monte Carlo tracking through the documented six-magnet geometry and apertures [6, 8]: source and transmitted velocity distributions and the acceptance function, all normalized; the dash-dotted line marks the measured mean. Script: `abs_acceptance_v1.m`.

3 The RHIC system

Throughout this note “RHIC” denotes the existing HJET holding-field magnet operated in the EIC hadron ring, i.e., with the EIC bunch structure of Table II of Ref. [1] ($f_b = 90.69$ MHz, $\sigma_t = 0.2$ ns, flattop beam). Present-day RHIC, with 9.4 MHz bunch spacing and 1.8 ns bunches, has an entirely different harmonic ladder and envelope, for which Sec. V of Ref. [1] finds no effect; nothing below refers to it — needed?

The RHIC holding field is produced by nested opposing coil pairs with iron core plates; the operating field is the difference of two larger single-circuit contributions. The field at the final installation was never measured (A. Zelenski, private communication, 2026); the value of 120 mT with a uniformity of 5×10^{-3} over the central 4 cm quoted in the 2006 thesis [9] restates the design calculation. → refer to Meng. — should be a reference

Our field model is the Opera-3d calculation of the magnet by W. Meng (2013), the only calculation of the magnet as designed and the basis on which its two currents were tuned at the time. What survives of it are three tables of the holding-field component along x , the horizontal direction toward the recoil detectors, through the magnet centre in the midplane, for three current settings; the model files are lost. Appendix B reproduces the tables, states their provenance, and derives from them the field along the atomic flight path: for an axisymmetric field free of sources near the axis, Laplace’s equation ties the axial profile to the midplane profile term by term, so that the midplane curvature fixes the axial curvature with the opposite sign and twice the magnitude, and the midplane fourth-order term fixes the axial fourth-order term. The result, at the experimental setting of 124.4 mT and scaled to the 120 mT setpoint, is

$$B(0, y, 0) - B(0, 0, 0) = a_y y^2 + a_{y4} y^4, \quad (16)$$

with $a_y = +3.42 \times 10^{-3}$ mT/mm² and $a_{y4} = -1.86 \times 10^{-6}$ mT/mm⁴, rising by 1.57 mT from the centre to the chamber boundary at ± 30 mm, with an uncertainty of a few tenths of a millitesla from terms the data do not constrain (Appendix B). The transverse variation follows from the same expansion. The coefficients scale with the setpoint, the magnet being set by current; Meng cautions that this holds only near the tabulated setting, since the iron saturates and the two circuits oppose each other, and the magnet as built reaches at most 124.8 mT with both circuits (199 mT with the inner circuit alone), so that the map of Fig. 6 above about 125 mT describes the field shape at an excitation this magnet cannot reach. The field at the final installation was

Also mention that RFL also has an RF shield in place and that ± 30 mm is its boundary.

ref A. Zelenski (priv. comm.)

Campaign can only be for $\mathcal{L}(x)$, not $\mathcal{L}(y)$

validation of earlier version

never measured; a measurement campaign at nominal currents is foreseen and will provide the first experimental characterization, and one traverse along the axis confirms the sign and size of the rise. Beyond due diligence this is a physics input, since unmodeled field structure along the path would add resonance traversals near the corridors and erode the valleys in a way the smooth model cannot show. Earlier versions of this note used a parabola with $a_{\eta} = +1.87 \times 10^{-3}$ and an excursion of 1.68 mT from a July fit whose script is lost; the excursion lies inside the present range, the coefficient was the effective parabola over the chamber region rather than the local curvature, and the origin of both is now on record.

The corridor widths depend on the axial excursion over the chamber region alone, and the results of this note do not depend on its sign or on its exact size. Figure 4 shows the RHIC map recomputed at production statistics for five variants of the static field: the profile of Eq. (16); the same profile with the sign of its deviation from the centre value reversed, which stands for the case that the axisymmetry assumption fails; the parabola of the earlier versions; and the first two with the drive region extended from the chamber boundary to ± 50 mm. Every 10^{-3} corridor edge is the same in all five to the 2 mT resolution of the scan (one edge, 56 against 58 mT, for the parabola); the floors at 120 mT span 1.07 to 1.23×10^{-4} , $\pm 7\%$ about the baseline; the optimum and the floors above 250 mT agree to 3%. Within the chamber region the edges are set by the on-axis dynamics and not by the 1.6 mT tilt of the profile toward the chamber boundary, and the extension of the drive region changes nothing because the profile does not carry an atom back through a crossing. The sign decides only on which side of each crossing the soft edge of Sec. 2.5 lies, below the resolution of the 10^{-3} contour.

needed

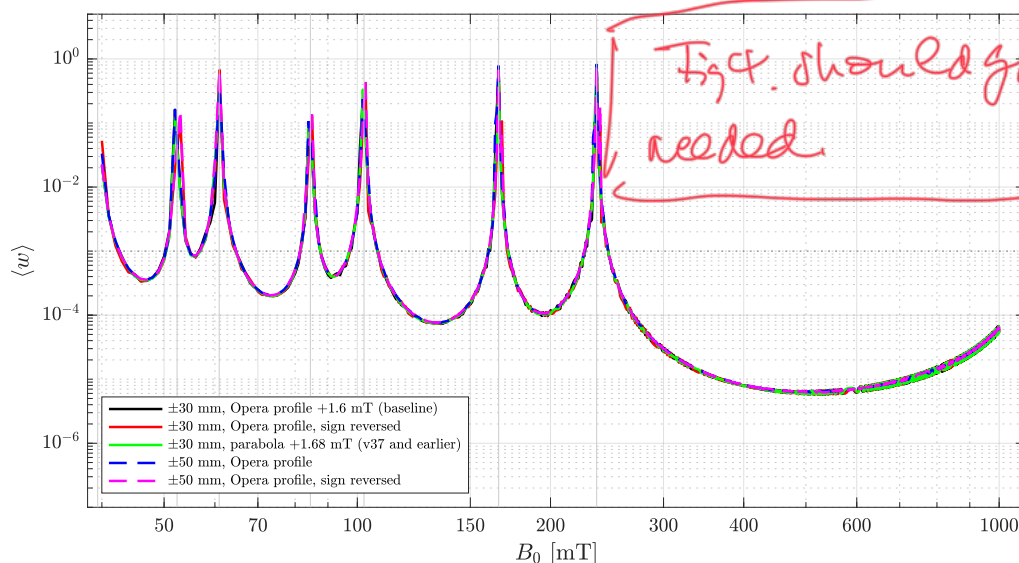
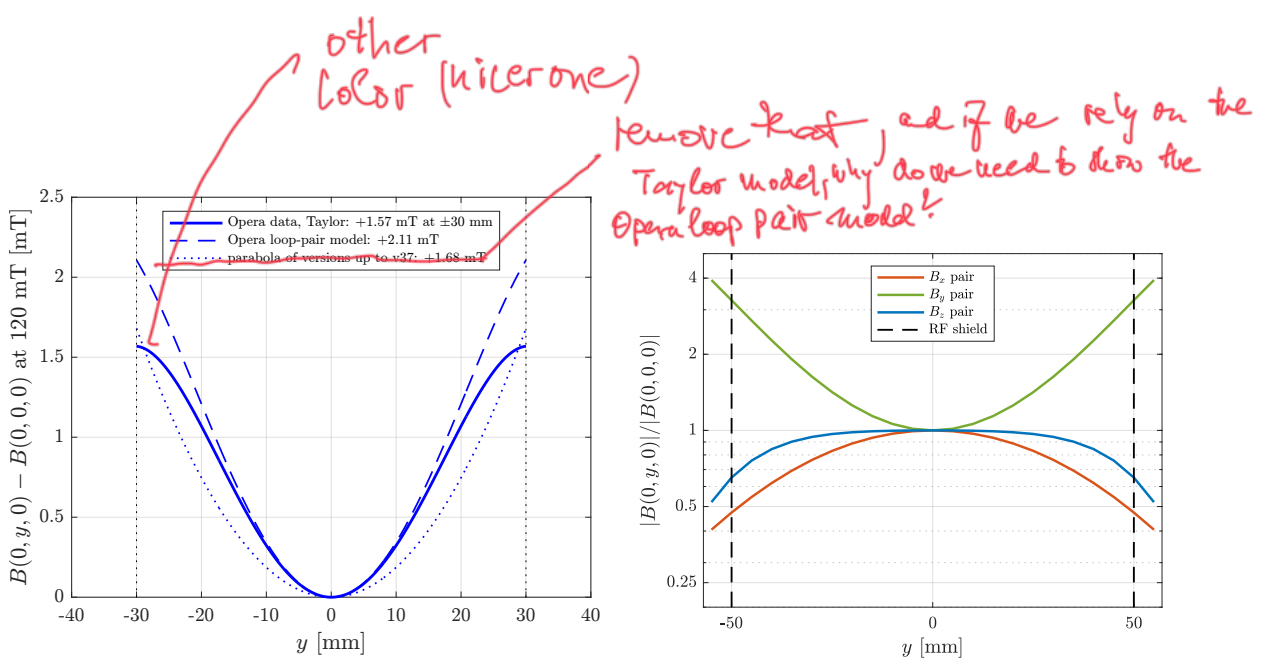


Figure 4: Sensitivity of the RHIC map to the static field model and to the drive-region boundary, production statistics: total transfer $\langle w \rangle$ for the profile of Eq. (16) (black), the same profile with reversed sign (red) and the parabola of the earlier versions (green), all with the chamber boundary ± 30 mm; and the first two with the drive region extended to ± 50 mm (dashed). Grey lines mark the crossings, the dotted line the 10^{-3} criterion. Scripts: `full_field_map_v29.m` (variants), `rhic_gate_compare_v9.m`.

Figure 5a shows the static profile along the flight path that sets the corridor width of Sec. 2.5. Figure 6 shows the resulting map. The 10^{-3} corridor edges lie at 98 to 108 mT, 160 to 172 mT and 226 to 246 mT around the f_{12} crossings at 102.4 and 236.0 mT and the f_{34} crossing at 166.1 mT, with the analogous fine corridors below 90 mT; above 247 mT an unconditional window extends to the f_{34} , $n = 8$ crossing at 1176.6 mT, the upper kinematic boundary of the elevated window

not needed in that detail figure is fully instructive, plus there is Table 2.



(a) RHIC magnet at the 120 mT setpoint, relative to the centre value: axial profile of Eq. (16) from the 2013 midplane tables (solid), loop-pair continuation of Appendix B (dashed), parabola of earlier versions (dotted); dotted verticals mark the chamber boundary.

(b) EIC HTS orientations at the 400 mT design excitation, $|B|$ normalized to the centre value, from the 3D field maps; dashed verticals mark the RF-shield boundary.

Figure 5: Static holding-field profiles along the atomic flight path y . Scripts: `opera_fit_v10.m` (a), `field_profiles_v6.m` (b).

(below, the 1177 mT boundary), whose tail produces the slow rise of the map beyond about 600 mT. At the operating point,

$$\frac{\Delta Q}{Q}(120 \text{ mT}) \approx 0.012\%, \quad (17)$$

robust against the atomic velocity, and in agreement at the few-percent level with the independent fixed-field evaluation of Ref. [2] (Eq. (84) therein), which validates the implementation. The operating point sits inside the valley between the 108 and 160 mT corridor edges, close to its local minimum, 7.3×10^{-5} at 135 mT (Table 2), within a factor 1.5 of the value at the historical operating point; a guide-field choice in the 120 to 130 mT range can hardly be an accident. The designers of the HJET evidently understood the resonance structure that, for lack of published record, is being rederived here, consistent with the statement in the 2006 thesis that the holding field was tuned to avoid beam-induced depolarization [9]. The total-transfer minimum of the landscape lies near 530 mT at $\langle w \rangle \approx 5.7 \times 10^{-6}$; $\langle w \rangle(400 \text{ mT}) = 7.9 \times 10^{-6}$.

4 The EIC system

The compact open geometry planned for the EIC serves three capabilities at once. First, statistics: six recoil-detector modules in three opposing pairs at 220 mm, replacing one pair at 750 mm, gain a factor 6.8 in the azimuthal figure of merit $F \propto \sum_i \Delta\phi_i \cos^2 \phi_i$, and together with the factor 3 from the EIC beam current the measurement time falls by about a factor 20 [1]. Second, physics reach: switchable target-field orientation gives principal access to all three beam-polarization components p_x , p_y , p_z , whereas the resistive RHIC magnet is fixed to the vertical and cannot be reoriented at all. Third, systematics: the open geometry admits a quadrupole mass analyzer inserted below the coil assembly, radial to the atomic beam, measuring the molecular-hydrogen fraction of the jet in situ, in dedicated intervals with the holding field reduced or switched off (QMA operation in the stray field of the coils is impractical), between fills or gated out of the polarimetry during stores; this closes the dilution systematic from nozzle-originating H_2 .

The RHIC system did not allow field reversal. add a picture with ded. f coils. say that coils are model.

K_2 is not coming from the nozzle!

can ρ_1 be mentioned in the caption

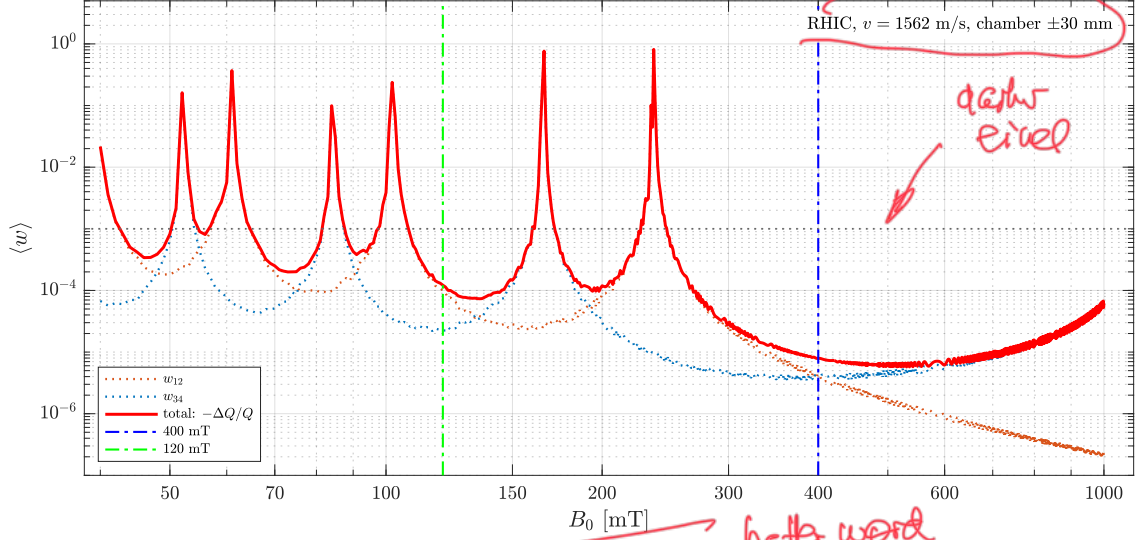


Figure 6: Ensemble-averaged transfer $\langle w \rangle$ versus the holding-field setpoint for the RHIC magnet; static field of Eq. (16), $v = 1562$ m/s, chamber drive region ± 30 mm, production statistics. Script: `full_field_map_v29.m`.

{ say that to go to min of $\langle w \rangle$, new magnet is needed }
[any way]

which is invisible to the profile subtraction and to the Breit–Rabi polarimeter alike (appendix of Ref. [1]; Sec. 3.4 of Ref. [2]), and is impossible with the RHIC magnet blocking every access path. The price is that the holding field in the 400 mT class must come from compact HTS coil pairs (square pairs for the transverse orientations, a circular near-Helmholtz pair for the ion-beam direction), whose limited size makes them intrinsically less uniform over the flight path; the present analysis establishes that each orientation possesses a viable operating window, i.e., that this price is payable. A cylindrical RF shield around the ion beam (inner radius ≈ 50 mm, with openings for the atomic beam) that confines the bunch fields to $|y| \lesssim 50$ mm along the atomic path.

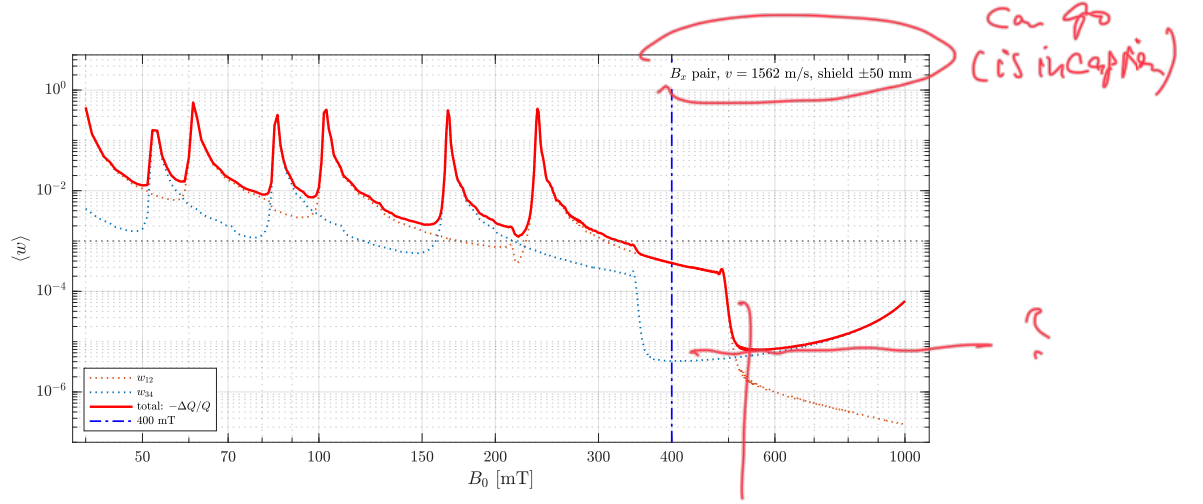
The static profiles along the atomic path, extracted from the 3D field maps at the 400 mT design excitation, are shown in Fig. 5(b); the excursions of $|B|$ are 5.8% (B_z pair), 23.6% (B_x) and 63.1% (B_y) over ± 30 mm, against 1.3% for the RHIC magnet; the corridor structure is correspondingly orientation dependent. The full trajectory dynamics, with $|B|$ interpolated from the 3D maps along every track and the drive gated by the shield, yields the maps of Figs. 7a to 7c and the operating-point summary of Table 2.

Table 2: Operating-point summary from the production maps; source `map_summary_v18.m`. Ranges are contiguous around the optimum; the optimum search is restricted to $B_0 \geq 260$ mT.

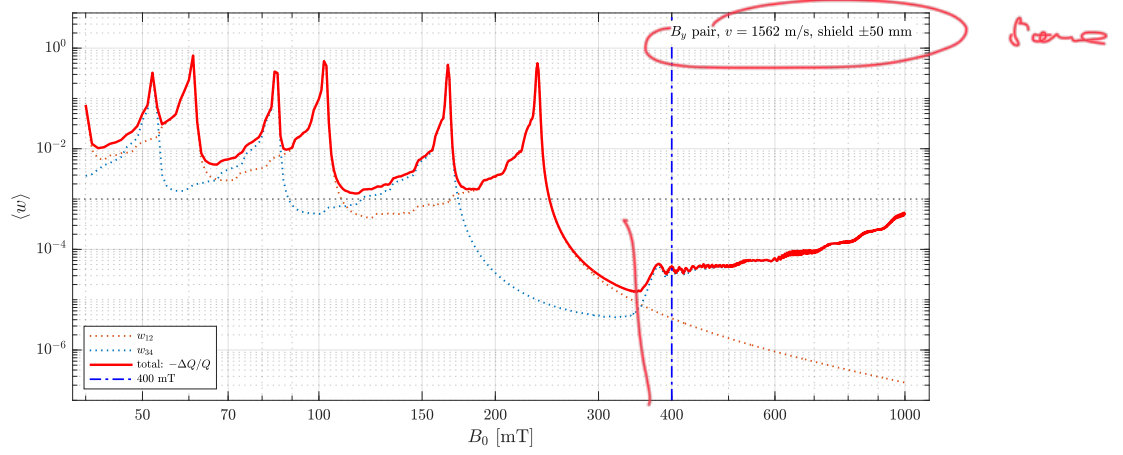
system	$\langle w \rangle (400 \text{ mT})$	B_{opt}	$\langle w \rangle (B_{\text{opt}})$	range $\langle w \rangle < 10^{-4}$
RHIC magnet	7.9×10^{-6}	531 mT	5.7×10^{-6}	268 to > 1000 mT
RHIC magnet, valley 108 to 160 mT	—	133 mT	7.4×10^{-5}	—
EIC B_z pair	8.5×10^{-6}	503 mT	6.3×10^{-6}	358 to > 1000 mT
EIC B_y pair	4.6×10^{-5}	349 mT	1.4×10^{-5}	271 to 713 mT
EIC B_x pair	3.6×10^{-4}	548 mT	6.7×10^{-6}	496 to > 1000 mT

The headline of Table 2: at 400 mT the compact B_z pair matches the RHIC magnet to within 8% (8.5×10^{-6} versus 7.9×10^{-6}), the justification of the HTS concept in one comparison. The B_y orientation is safe at 400 mT (4.6×10^{-5}) but prefers the lower window (optimum 349 mT) and is ceiling-limited at the 10^{-4} level at 713 mT: its field rises steeply along the path toward the coils,

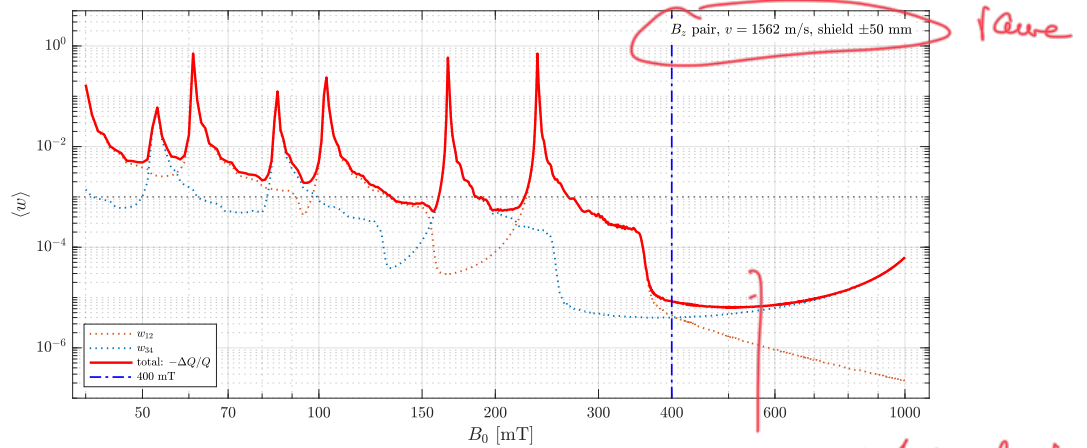
?



(a) B_x orientation.



(b) B_y orientation; the rise above 500 mT reflects the field swept along the path approaching the f_{34} , $n = 8$ crossing at 1177 mT, the upper kinematic boundary of the elevated window.



(c) B_z orientation; the cleanest of the three.

not good, no interpretation of this look

Figure 7: EIC orientation maps: $\langle w \rangle(B_0)$ at $v = 1562$ m/s, RF shield ± 50 mm, production statistics. Script: full_field_map_v29.m.

and above roughly 500 mT the field swept along the path approaches the 1177 mT boundary from below, the atoms traversing the associated crossing inside the shield where the drive is $1/r$ -suppressed but nonzero (the resonance-traversal regime of Sec. 2.4). The B_x orientation carries its 236 mT-corridor tail to 496 mT and is clean above. Its floors are conservative by the factor 2.6 of Sec. 2.7, the same as for the B_y pair; the suppression of the on-axis coupling by two orders of magnitude, which one might expect from the azimuthal bunch field being parallel to B_x on the axis, does not survive the weighting by the beam density (Eq. (13)), since the atoms that scatter sit off axis inside the beam. Below the elevated window the orientations differ: B_x and B_y exclude the low-field region outright (10^{-3} contour edges at 325 and 248 mT), while B_z retains two narrow islands, 132 to 160 and 186 to 224 mT, at the 10^{-3} criterion, none of which survives at 10^{-4} . The common low-field window across the three orientations is empty; for the EIC system as a whole the elevated-field window is mandatory. Within it the design fork is quantitative: either a common setpoint near 500 to 550 mT (inside the 10^{-4} intersection 496 to 713 mT of all three orientations), or per-orientation setpoints (B_z 503, B_y 349, B_x 548 mT) at 6.3×10^{-6} to 1.4×10^{-5} each; which is operable is a question for the polarimetry side. The staged detector plan makes the choice natural for the first phase: Phase 1 operates the horizontal detector pair with the B_y coils alone, so no common-setpoint compromise exists yet and Phase 1 can run at the B_y optimum near 349 mT; the common-versus-individual question arises with the added orientations of Phase 2.

too much better

5 Inputs and sensitivities

not a good title - systematic effects ... is better

Table 3 states the inputs of the analysis and the effects included, estimated or not yet included; it is the validity statement of the results above. The impact vocabulary of the table: exact (fixed by the model), percent-level (moves quoted values within their last digits), spec item (requirement on operations), none-kinematic (no crossing exists in the window at any harmonic), check required (open until the stated check exists).

I think that the table 3 is excessive.

Discuss: The core model inputs should go to a separate table

• Lines between items

• Need to go through what is needed for follow-up paper.

• There will be follow-up paper on systematics of the Q determination using BRP.

• Looking at this, I would find a paper identifying the errors for the new gentle high system sufficient. Need and it justifies a paper.

• Systematics is discussed here in yet another paper.

• we could add a lot of effects that need to be investigated.

Table 3: Inputs and effects, grouped by type, with the impact at the elevated-field operating point (vocabulary defined in the text). Status: I = included, E = estimated only, N = not included.

input / effect	value or source	impact at operating point	status
CORE MODEL INPUTS			
bunch envelope, duty factor	Table II of [1]	exact within model	I ✓
Rabi prefactor, Fourier factor	Eqs. (6), (7); checked against Eq. (84) of [2]	exact within model, validated	I
Breit–Rabi matrix elements	Eq. (3)	exact	I
drive polarization geometry	bunch field taken fully transverse for all orientations; beam-weighted projection computed (Sec. 2.7)	conservative: floors $\times 0.38$ (B_x , B_y , RHIC), $\times 0.76$ (B_z) for the polarimetry; edges exact	E ??
bunch length σ_t	0.200 ns per Table II of [1]	exponential input: $\sigma_t = 0.3$ ns lowers the $n = 7, 8$ amplitudes by 4 to 5, weakening the 236 mT corridor and the 1177 mT boundary; longer bunches only help	I
bunch intensity, beam size	$N_p = 6.9 \times 10^{10}$, $\sigma_r = 0.656$ mm, one beam	floors $\propto N_p^2$; two beams through the jet would double the drive (question to operations)	E ?
beam-field radial profile	Eq. (4), peak 3.0 mT	exact within model	I
bunch spacing	11.027 ns per Table II of [1]	sets all crossing fields; RF-bucket sensitivity noted	I
FIELD REALIZATION			
static field model	RHIC: 2013 Opera midplane tables (Meng), axial profile by the harmonic relation, Eq. (16), Appendix B; EIC: 3D maps	corridor positions $\sim 1\%$; edges insensitive to sign and size of the axial rise (Fig. 4), floors $\pm 7\%$; field campaign confirms	I
setpoint realization	operating field is the difference of two large opposing coil contributions; supply errors amplified by their ratio; the same balance sets the axial curvature	dominant operational item	E
field re-landing after orientation switches and H_2 intervals	three-axis sequencing, $\sim 10^3$ cycles per store	must stay clear of the corridor edges; spec ± 10 mT	E
RF shield boundary	≈ 50 mm, drawing pending	sets B_y/B_x drive exposure	E

Table 3: Continued.

input / effect	value or source	impact at operating point	status
TARGET ENSEMBLE			
atomic velocity, mean	1562 ± 50 m/s measured [5]	percent-level via traversal regime	I
velocity spread	12% rms computed, cutoff < 1300 m/s	percent-level; B_y most sensitive	I
trajectory phase space	transverse offsets ($\sigma = 2.6$ mm) and inclinations (20 mrad rms) drawn independently; position–angle correlations await the ABS-tracked ensemble	percent-level on quoted values	E
BEAM FINE STRUCTURE			
full harmonic comb per crossing	resonant harmonic only	none in elevated window (kinematic)	N
abort gap, revolution sidebands	$1 \mu\text{s}$ published for RHIC and EIC; sideband comb estimated	none in elevated window (kinematic); valley floors carrier-only, waveform bound in preparation	E
bunch-to-bunch intensity ripple	$\sim 10^{-3}$ per revolution line	none in elevated window (kinematic)	E
bunch-length spread	10% spread raises $n \approx 40$ amplitudes $\times 5$ to $\times 10$	none in elevated window (kinematic)	E
non-Gaussian longitudinal profile	dual-harmonic RHIC stores documented; EIC store structure open	none in elevated window (kinematic)	N
elliptical beam	$\sigma_x = 1.61$, $\sigma_y = 0.27$ mm at flattop; field inside the beam linear, $\langle B_x^2 \rangle = \langle B_y^2 \rangle$ for any aspect ratio (Eq. (13))	contained in the beam-weighted factors above; no orientation favoured	E
ENVIRONMENT AND OBSERVABLE			
shield cavity modes	mode spectrum vs. transition frequencies	design check required	N
coherence terms in ΔQ	bunch-overlap convolution	systematics analysis	N
BRP transmission of transferred atoms	with the ABS-simulation effort	systematics analysis	N
mechanical vibrations (cryopumps 1 Hz, turbos 0.5 to 1.5 kHz)	sextupole lenses move by tens of μm , beam centroid sways ≤ 0.1 mm; spin follows adiabatically (rate ratio 10^{-10}); sway frozen during a $38 \mu\text{s}$ transit, sidebands inside the 26 kHz transit width	none: intensity modulation only; BRP switching cycle to avoid the displacer frequency (systematics analysis)	E

Three sensitivities deserve emphasis. First, all spectral uncertainties of the beam (satellite bunches of the documented dual-harmonic RHIC stores, abort-gap revolution sidebands, intensity ripple, bunch-length spread) attack the highly suppressed high-harmonic transitions and the low-field point estimates; none of them touches the elevated window, whose safety is kinematic: no crossing of f_{12} or f_{34} exists there at any harmonic, so no spectral component, known or unknown, can drive a resonance. The same protection does not extend to operation inside a valley, such as the RHIC point at 120 mT: there the quoted floor is computed against the resonant carrier alone, while the abort gap dresses every harmonic with revolution sidebands at ~ 78 kHz spacing and $\sim 8\%$ relative amplitude, so the nearest sideband, not the carrier, sets the smallest detuning; a waveform-level bound, integrating the gapped fill pattern, is in preparation, and until it exists the 120 mT floor carries this caveat. Second, all crossing fields scale with the bunch frequency; the analysis uses $\tau_b = 11.027$ ns of Table II of Ref. [1] throughout, and the sensitivity to the final EIC RF-bucket configuration is noted. Third, the RF shield forms a

1. explain better

and also for one for the EIC

cavity whose mode spectrum should be checked against the transition frequencies at the chosen operating field; this is a one-line design requirement. Fourth, the three-axis sequencing and the H₂-measurement intervals ramp the field through the exclusion corridors and the low-field region on every switch; this is harmless, because the jet has no memory, atoms transit in tens of microseconds and are continuously replenished, so corridor traversal during a switch depolarizes only the atoms of that instant and biases no measurement, while the setpoint reproducibility after each switch is a genuine specification item (Table 3). The H₂ measurement itself assumes the molecular fraction to be field independent, which is physically expected (neither the surface recombination nor the nozzle dissociation depends on fields at the sub-tesla scale) but is stated here as an assumption.

6 Summary and outlook

With the corrected harmonic amplitudes, beam-induced depolarization of the HJET at the EIC is governed by the two transitions f_{12} and f_{34} alone, and by the interplay of their resonance kinematics with the static field distribution along the atomic trajectories. At the RHIC operating point the corrected effect is $\Delta Q/Q \approx 0.011$ to 0.012% , velocity-robust and consistent with the independent evaluation of Ref. [2]. For the EIC, the compact HTS system with its RF shield sustains RHIC-grade performance in the B_z orientation at 400 mT, the elevated-field window is mandatory for the system as a whole, and the per-orientation optima and limits of Table 2 constitute direct design input, including a quantitative fork between a common setpoint near 500 to 550 mT and per-orientation setpoints.

Two items of the RHIC magnet remain open as facts, not as risks: the axial profile, derived here from the 2013 midplane calculation under the assumption of axisymmetry and confirmed by one on-axis traverse of the foreseen measurement campaign, and the real field structure along the path, which the same campaign will provide.

This note is the analysis record. A minimal Erratum to Ref. [1] is in preparation on its basis (Appendix C); the bulk of the material is intended for a dedicated publication.

7 Acknowledgement

W. Meng (formerly BNL) provided the 2013 field tables of the RHIC holding-field magnet and their description. R. Porqueddu (BNL) provided a finite-element field map of a coil-and-iron system used to check the midplane-to-axis relation of Appendix B.

A Reproduction

All figures and tables of this note are generated by the versioned scripts of Table 4 and carry the generating script and parameters in their captions and filenames; a figure can therefore never silently outlive the code state that produced it. The pipeline `run_all.v30.m` executes the full set in sequence after four yes/no questions (EIC maps, production quality, RHIC main map, sensitivity variants), preserves all figures on screen, prints the wall-clock time per step, and writes the complete console output to a timestamped diary file. The production maps of this version are those of `runlog_2026-09-04.1951.txt` (six hours with all options); the remaining figures are from `runlog_2026-09-05` of the same pipeline. The version history of this note is maintained in the accompanying file `technote_changelog.v40.txt`.

Table 4: Analysis scripts, current versions, and their outputs.

script	output
opera_fit.v10.m	RHIC field model from the 2013 tables (Appendix B)
field_profiles.v6.m	EIC static profiles (Fig. 5(b))
abs_acceptance.v1.m	velocity acceptance (Fig. 3)
lowharm_passage.v10.m	single-crossing dynamics (Fig. 1)
clearance_scan.v7.m	clearance at one crossing (Fig. 2)
full_field_map.v29.m (production)	Figs. 6, 7a to 7c, variants of Fig. 4
full_field_map.v31.m	current version, identical physics, reads the v10 cache
map_restyle.v15.m	restyles map figures from saved results
map_summary.v18.m	Table 2
beam_weighting.v1.m	beam-weighted drive factors (Table 1)
rhic_gate_compare.v9.m	sensitivity figure (Fig. 4)
run_all.v30.m	pipeline, timing and diary log

B The holding field of the RHIC magnet from the 2013 Opera tables

B.1 Provenance

The holding-field magnet of the HJET consists of two nested pairs of coils on a common vertical axis, inner pair $N = 56$ turns, outer pair $N = 40$ turns, powered in opposition and fitted with iron plates; the operating field is the difference of two larger single-circuit contributions. The original design goal was a field-free region for the recoil protons; in practice the ratio of the two currents was tuned so that the field integral along the recoil-proton path through the magnet becomes small (W. Meng, private communication, 2024). The magnet was modelled in Opera-3d by W. Meng, who provided field tables for that tuning; the last delivery, on 13 May 2013, comprises three tables at inner and outer currents of (348, 275), (349, 276) and (349, 275) A, the last being the experimental setting. Each table gives the holding-field component along the Opera x axis from the magnet centre to 800 mm in 2 mm steps and closes with the field integral along that line, 594, 583 and 641 G cm. The x axis is the horizontal direction toward the recoil detectors, perpendicular to the ion beam, in the midplane: the field integral along it, the one seen by the recoil protons, is the quantity the tables were made for. No table along the vertical axis exists, and the Opera model files no longer exist (Meng, 2024). The magnet as built reaches at most 1248 G with both circuits at 353.5 and 292 A, and 1991 G with the inner circuit alone (Meng’s notebooks); Meng cautions that the field cannot be scaled with current beyond the tabulated settings because of the iron and the opposing circuits. Table 5 reproduces the three tables condensed; the complete tables accompany this note as supplementary files.

B.2 What the tables give directly

The three centre fields, 123.829, 124.128 and 124.400 mT, determine the two circuits’ contributions by least squares,

$$B_0 = c_{\text{in}} I_{\text{in}} - c_{\text{out}} I_{\text{out}}, \quad c_{\text{in}} = 0.5705 \text{ mT/A}, \quad c_{\text{out}} = 0.2716 \text{ mT/A}, \quad (18)$$

with residuals below 10^{-4} mT; at the experimental setting the field is $199.1 - 74.7 = 124.4$ mT. The three profiles have the same shape to four digits (Fig. 8), so the shape analysis uses the experimental setting alone. Along x the field falls from the centre, by 2.12 mT (1.70%) at 30 mm, crosses zero near 100 mm and reaches -190 mT at 150 mm, the return flux through the iron. Near the centre the profile is even and smooth; even-polynomial fits over $|x| \leq 20$ mm represent

Table 5: The 2013 Opera tables of W. Meng, condensed: holding-field component B_y along x , toward the recoil detectors, through the magnet centre in the midplane, for the three current settings (inner, outer) in amperes. Original units gauss and centimetres; 2 mm steps to 800 mm in the full tables.

x [mm]	B_y (348, 275) [mT]	B_y (349, 276) [mT]	B_y (349, 275) [mT]
0	123.829	124.128	124.400
2	123.821	124.120	124.392
4	123.801	124.099	124.371
6	123.765	124.063	124.335
8	123.713	124.011	124.283
10	123.645	123.943	124.215
12	123.559	123.857	124.129
14	123.454	123.752	124.024
16	123.328	123.625	123.898
18	123.180	123.476	123.750
20	123.005	123.301	123.575
22	122.802	123.097	123.372
24	122.569	122.864	123.139
26	122.291	122.585	122.860
28	121.997	122.289	122.566
30	121.714	122.006	122.283
40	119.155	119.439	119.720
50	114.582	114.851	115.138
60	106.797	107.042	107.338
80	74.083	74.228	74.552
100	9.307	9.255	9.631
120	-84.366	-84.707	-84.239
150	-186.434	-187.090	-186.512
200	-47.979	-48.136	-48.043
250	16.832	16.901	16.801
300	14.784	14.842	14.767
400	6.038	6.062	6.032
500	2.966	2.978	2.963
600	1.692	1.698	1.690
800	0.740	0.742	0.739

it to 2×10^{-4} mT and give the midplane coefficients

$$B(x, 0, 0) = B_0 + a_r x^2 + a_{r4} x^4, \quad a_r = -1.772 \times 10^{-3} \text{ mT/mm}^2, \quad a_{r4} = -7.24 \times 10^{-7} \text{ mT/mm}^4, \quad (19)$$

at 124.4 mT (Fig. 9, Table 6).

B.3 From the midplane to the axis

The field along the atomic path is the axial profile $f(y) \equiv B_y(0, y, 0)$, for which no table exists. It follows from the midplane profile because the field near the axis is free of sources and, to the accuracy needed, axisymmetric: the coils are coaxial, and the iron plates and the chamber cut-outs are far compared with the 20 mm the analysis uses. In a source-free region $\vec{B} = -\nabla\Phi$ with $\nabla^2\Phi = 0$, and the axial component $B_y = -\partial\Phi/\partial y$ is itself harmonic. In cylindrical coordinates (r, φ, y) about the axis, with no φ dependence,

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial B_y}{\partial r} \right) + \frac{\partial^2 B_y}{\partial y^2} = 0. \quad (20)$$

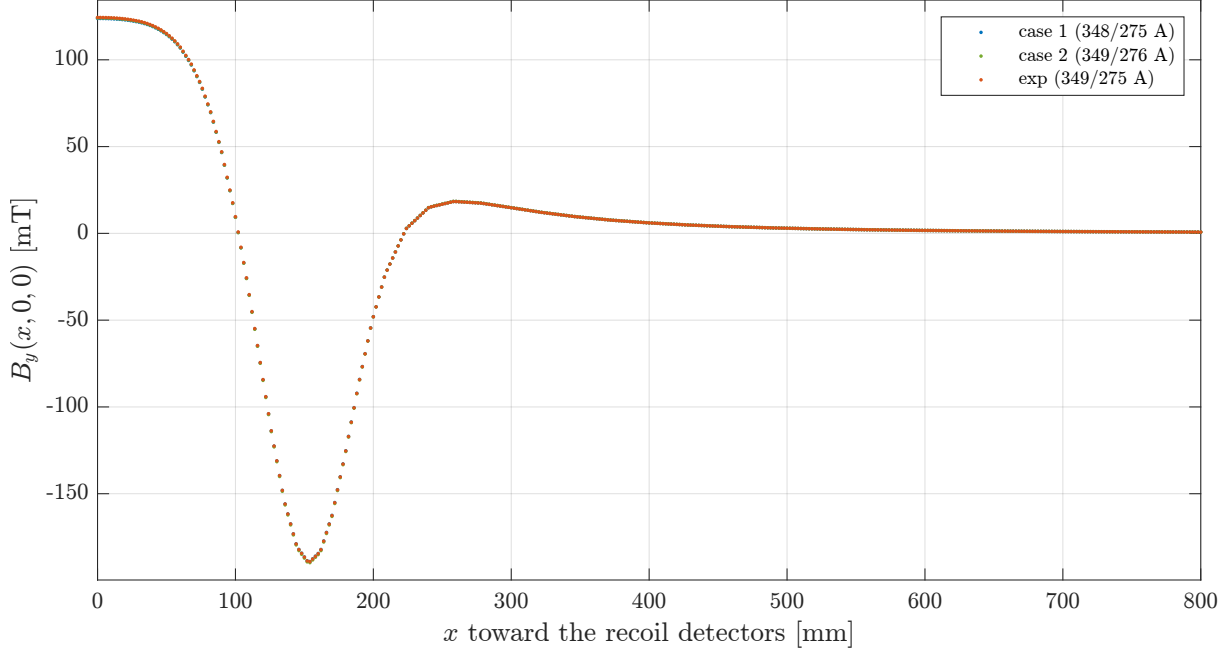


Figure 8: The three 2013 Opera tables over their full range, holding field along x , toward the recoil detectors. Script: `opera_fit_v10.m`.

480 Regularity on the axis allows only even powers of r ; inserting $B_y(r, y) = \sum_{n \geq 0} c_n(y) r^{2n}$ with
 481 $c_0 = f$ gives the recursion $(2n)^2 c_n + c''_{n-1} = 0$, hence

$$B_y(r, y) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{r}{2}\right)^{2n} f^{(2n)}(y) = f - \frac{r^2}{4} f'' + \frac{r^4}{64} f'''' - \frac{r^6}{2304} f^{(6)} + \dots, \quad (21)$$

482 the classical expansion of an axisymmetric field about its axis. Evaluated in the midplane,
 483 $y = 0$, with $f(y) = B_0 + a_y y^2 + a_{y4} y^4 + a_{y6} y^6 + \dots$, so that $f''(0) = 2a_y$, $f''''(0) = 24a_{y4}$ and
 484 $f^{(6)}(0) = 720a_{y6}$, the coefficients of Eq. (19) are

$$a_r = -\frac{a_y}{2}, \quad a_{r4} = \frac{3}{8} a_{y4}, \quad a_{r6} = -\frac{5}{16} a_{y6}, \quad (22)$$

485 and inverting,

$$a_y = -2a_r, \quad a_{y4} = \frac{8}{3} a_{r4}, \quad a_{y6} = -\frac{16}{5} a_{r6}. \quad (23)$$

486 A field that falls quadratically along x must rise quadratically along the axis, with twice the
 487 curvature: the centre of a magnet whose field is maximal in one direction is a saddle point, and
 488 the two transverse directions share the compensating curvature between them.

489 The relation was checked on a finite-element field map of a coil-and-iron system for which
 490 both directions were available (a $67 \times 223 \times 67$ export at 0.45 mm, R. Porqueddu, 2026): its
 491 midplane ring means give $a_r = +4.65 \times 10^{-4}$ and $a_{r4} = -6.34 \times 10^{-7}$, from which Eq. (23)
 492 predicts axial coefficients $a_y = -9.31 \times 10^{-4}$ and $a_{y4} = -1.69 \times 10^{-6}$; the map's own axis gives
 493 -9.79×10^{-4} and -1.51×10^{-6} , and the predicted excursion at 30 mm, -2.21 mT, against
 494 -2.09 mT read from the map. The relation holds to 6% where it is applied.

495 B.4 Result and uncertainty

496 Table 6 applies Eq. (23) to even-polynomial fits of the experimental-setting table for four windows
 497 and two orders. Where the fit represents the data to their precision, windows up to 20 mm with
 498 rms 2 to 3×10^{-4} mT, the coefficients are stable: $a_y = +3.52$ to 3.54×10^{-3} mT/mm², $a_{y4} = -1.9$

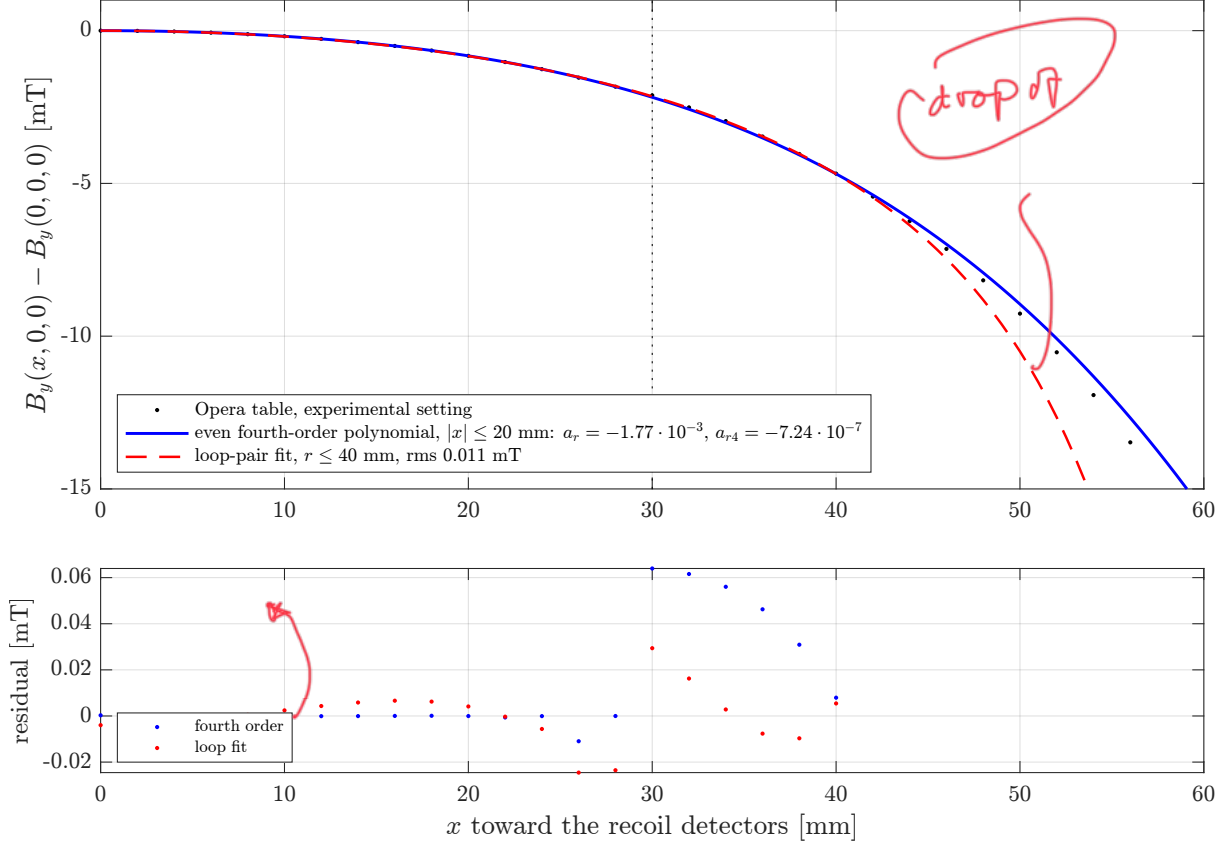


Figure 9: The experimental-setting table near the centre, relative to the centre value, with the even fourth-order polynomial of Eq. (19) fitted over $|x| \leq 20$ mm and the three-loop-pair continuation fitted over $r \leq 40$ mm; the lower panel shows the residuals of both. The dotted line marks the chamber boundary. Script: `opera_fit_v10.m`.

to -2.0×10^{-6} mT/mm⁴, and the axial rise over the chamber region, $B(0, \pm 30, 0) - B(0, 0, 0)$, is +1.55 to +1.63 mT at 124.4 mT. For windows of 30 and 40 mm the polynomial no longer represents the data (rms 0.01 to 0.02 mT, the iron making itself felt) and the y^6 term, multiplied by 30⁶, swings the rise anywhere between -1 and $+3$ mT: the data within 20 mm fix a_y and a_{y4} , and with them the axial profile out to about 20 mm; at 30 mm the rise carries an extrapolation uncertainty of a few tenths of a millitesla from the terms the data do not constrain. We adopt the 20 mm, fourth-order values,

$$a_y = +3.544 \times 10^{-3} \text{ mT/mm}^2, \quad a_{y4} = -1.930 \times 10^{-6} \text{ mT/mm}^4, \quad (24)$$

giving $B(0, \pm 30, 0) - B(0, 0, 0) = +1.63$ mT (+1.31%) at 124.4 mT, which scaled to the 120 mT setpoint are the values of Eq. (16) in Sec. 3, and quote the rise as +1.6 mT with an uncertainty of ± 0.3 mT.

As an independent continuation, a model of three coaxial current-loop pairs (radius, half-gap and ampere-turns free; exact circular-loop fields) was fitted to the table over $r \leq 40$ mm: rms 0.011 mT, pairs of radii 95, 122 and 71 mm at half-gaps 57, 44 and 14 mm. The loop model is harmonic by construction, satisfies Eq. (22) to 2%, and continues the axial profile to +2.18 mT at 30 mm; fitted over $r \leq 100$ mm it degenerates (rms 0.10 mT) and gives +1.41 mT. The loop model is a representation, not a model of the magnet: it does not reproduce the return flux, and it is shown in Fig. 5a only to bracket the continuation beyond 20 mm.

Earlier versions of this note carried $a_y = +1.867 \times 10^{-3}$ and an excursion of 1.68 mT from a July 2026 loop-pair fit whose script is lost, together with $a_r = -1.78 \times 10^{-3}$, and noted that the pair violates Eq. (22). The present analysis resolves this: a_r was the midplane curvature of the

Table 6: Midplane even-polynomial fits of the experimental-setting table and the axial coefficients that follow from Eq. (23); rise = $B(0, 30, 0) - B(0, 0, 0)$ at 124.4 mT. Coefficients in mT/mmⁿ, in the units given in the column heads. Source `opera_fit_v10.m`, section T.

W [mm]	order	a_r [10^{-3}]	a_{r4} [10^{-7}]	a_{r6} [10^{-10}]	a_y [10^{-3}]	a_{y4} [10^{-6}]	rise [mT]	rms [mT]
13	4	-1.768	-7.56	—	+3.535	-2.02	+1.55	0.0003
20	4	-1.772	-7.24	—	+3.544	-1.93	+1.63	0.0002
20	6	-1.772	-7.23	-0.00	+3.544	-1.93	+1.63	0.0002
30	4	-1.852	-5.93	—	+3.704	-1.58	+2.05	0.0108
30	6	-1.643	-12.7	+5.2	+3.285	-3.38	-1.00	0.0067
40	4	-1.723	-7.38	—	+3.445	-1.97	+1.51	0.0175
40	6	-1.903	-4.09	-1.5	+3.806	-1.09	+2.88	0.0118

table over a small window; the excursion 1.68 mT is inside the present range; and the quoted a_y was the effective parabola coefficient over the chamber region ($1.68/900 = 1.87 \times 10^{-3}$), not the local curvature, which the fourth-order term reduces from 3.2 mT (parabola) to 1.6 mT at 30 mm.

The static field used in the trajectory code is

$$B(x, y, z) = \frac{B_0}{f(0)} \left[f(y) - \frac{x^2 + z^2}{4} f''(y) \right], \quad f(y) = f(0) + a_y y^2 + a_{y4} y^4, \quad (25)$$

the first two terms of Eq. (21) with the axial polynomial of Eq. (24), scaled from 124.4 mT to the setpoint B_0 ; the transverse term contributes below 0.05 mT for the ensemble of Sec. 2.8. The profile is written by `opera_fit_v10.m` to the file `opera_onaxis_v10.txt`. The map script (`full_field_map_v31.m`) reads it; the production maps of this version were computed with version 29 of that script on the identical profile.

B.5 Caveats

Three assumptions enter. Axisymmetry near the axis: the iron plates and the chamber cut-outs break it at the 100 mm scale; the analysis uses data within 20 mm, and the measurement campaign, with one traverse along the axis, tests the result directly. Linearity with current: the coefficients are used at the tabulated setting and scaled with the setpoint for the map; Meng's caution applies above the tabulated excitation, and the magnet cannot in any case exceed 125 mT with both circuits, so the RHIC map above that field describes the field shape at an excitation this magnet does not reach. Finally, the tables are a calculation, not a measurement: the field at the installation was never measured. An independent finite-element reconstruction of the magnet, made in 2026 without the drawings, gives a different sign of the axial curvature; the comparison of the loop-pair fits shows the two to differ in the effective coil radii by 40%, which decides the sign at the level of the as-built dimensions that nobody has. It is not used here. Figure 4 shows that neither the sign nor the size of the rise within this range affects the results of the note.

C Draft of the Erratum

This appendix is the single working draft of the Erratum to Ref. [1]; it is iterated only here and leaves this note as the submitted document. All numbers are from the production-quality runs; bold brackets mark the two items fixable only at submission, the final equation numbering against the journal layout and the AI-assistance disclosure.

[Author line of the Erratum: the full author list of Ref. [1]; the single-author line of this note does not extend to the Erratum.]

Erratum: Eliminating beam-induced depolarizing effects in the hydrogen jet target for high-precision proton beam polarimetry at the electron-ion collider [Phys. Rev. Accel. Beams 29, 021001 (2026)]

The spectral amplitudes shown in Figs. 11 and 16 of the paper represent the single-bunch spectral envelope normalized to the peak field and omit the Fourier normalization of the periodic bunch train. The correct full oscillation amplitude of harmonic n of the beam-induced field is

$$B_1^{(n)}(r) = 2\sqrt{2\pi}\sigma_t f_b B_{\text{pk}}(r) e^{-2\pi^2 n^2 f_b^2 \sigma_t^2},$$

smaller than the amplitudes used in Sec. VI.D by the duty factor $\sqrt{2\pi}\sigma_t f_b \approx 1/22$ and, for the quoted Rabi frequencies, by a factor of about 60 in total. As a consequence, the example of Sec. VI.D based on the spatial field amplitude is not tenable, and the statement that operation at the RHIC holding field of 120 mT would be untenable at the EIC is withdrawn. The corrected evaluation gives $|\Delta Q/Q| \approx 0.012\%$ at 120 mT for the nominal EIC flattop beam, negligible at the required level and consistent with the independent evaluation of Ref. [2].

Appendix C is corrected as follows: the ensemble-averaged transition probability is $\bar{\Pi} = (\pi/2)\omega_R^2 \tau_{\text{int}} \rho(0)$, with ρ the detuning density along the trajectory; this replaces Eqs. (C2) and (C3), which double-counted the interaction volume and omitted the time dependence.

Unaffected are Secs. III and IV, Figs. 6, 9, 14 and 17, Table IV, and Appendices A and B, whose content does not depend on the harmonic normalization. The choice of the operating field involves considerations beyond beam-induced depolarization and is addressed in a dedicated forthcoming analysis.

[AI-assistance disclosure per the APS policy to be included at submission.]

) drop this ¶?

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